Analysis

and

Design

Senior Project by: Christopher Corbett

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1 December 2001

Jody Jackson
7006 Capital Way,
Springfield, SC 29146

Subject: Complete Analysis

Dear Jody Jackson:

Mr. Jackson, I have finally finished my in-depth analysis of your system. By implementation all of my suggestions you and your company should be able to increase your efficiency and accuracy in regards to your pump reconciliation's during the next several years.

I would like to encourage you to call me should there be any questions, however I feel that the enclosed report should be adequate to answer most any question you may have. The system design should be implemented within the next few months to ensure that your system is fully functional by the next fiscal quarter.

The following report contains all of my findings, and should give you the right idea about your system upgrade requirements.

Thank you for the pleasure of working with you and for you—and I hope that in a few yours when your system is again in need of an upgrade, you will not hesitate to call me.

Sincerely,

Cc

System Requirements:

The system to be installed and implemented is the one found under the Candidate Matrix. It will use both Microsoft Access 2000 and Microsoft Visual Basic 2000, so both of those programs will have to be purchased and installed. I would suggest that at a minimum, your system have these specifications:

- Windows 2000
- 64 Meg. RAM
- .5 Gig of Hard drive space. (If both Access and VB are already installed, only require 20 Mb)
- CD-ROM (12x or greater)

Installation Instructions:

The program will be presented on a CD.

1. Open up Explorer, and create the following directory. <c:\Program Files\Pumps >
2. Go to Start → Run → D:\Setup.exe **(or)**
   a. Start → Run → Browse
   b. An "Open" window will open and allow you to select the Setup.exe program from the CD.
3. After it is installed, you can load it from either the icon on the desktop, or from Start → Programs → Pumps → Reconcile (Name of the program)

Purpose

The Capital Way Food Mart in Springfield, SC uses a very outdated and inefficient system of carrying out all of its pump reconciliations from day to day. This design will enable the store to automate their system—allowing them to keep better records.

Background

Capital Way Food Mart has been doing their pump and tank reconciliations in much the same way for around 15 years. Their system is outdated, and can not expect to keep up with a changing market demanding perfection in paperwork.

To date, the only way that the Texaco gas station has to reconcile its pumps is to fill out a form provided to them by Texaco. The form ultimately keeps track of the gas that has passed through the tanks, and allows for checking for whether or not there is a leak, or perhaps water in the system. Unfortunately, this form is prone to a variety of errors and can be fraudulently manipulated.

I. Executive Summary

A. Summary of Recommendations

In the course of my analysis, I have found that the system in use is inefficient in carrying out everyday operations in a small town gas station. The employees are expected to—at the end of their day—go out and read the total gallons sold off of the pumps. They then calculate any discrepancies between their gallons sold, and the gallons that have actually passed through the pumps; figuring out whether or not there have been any drive offs or if someone did not ring up a gas sale. They will also determine if the projected gallons left in the tanks is within a tolerance of how many gallons there actually are. To do this, the employees will calculate the remaining gallons in the tank by adding up the total amount sold since the last refilling. Then, someone will go out and take a stick reading of how many inches are left in the tank. If both readings are not within a tolerance of each other, than there is either a pump that is not reading correctly—or more likely a leak in one of the tanks.

I would recommend that that a program be designed to help the employees to do their pump reconciliations. That system would enable the store to have the reconciliations completed accurately—cutting out a lot of human error and also making the records more efficient and easier to review.

B. Brief statement of anticipated benefits

The software updates will allow this company to better keep track of its fuel in each one of the several tanks. There are tanks for every grade of gas, as well as a tank for desiel [on and off road], and one for kerosene. It would also be nearly impossible to alter any of the records in the future to hide mistakes in the past. The files could be encrypted—making them impossible to adjust. Furthermore, they would be more accurate and much easier to keep up with. The manager would only have to worry about a diskette that is easily backed up and stored, rather than hundreds of papers over time.

C. Brief explanation of report contents

The rest of this report will contain background information about this project, which will include a brief description of the project requests, and a brief explanation of the survey phase activities. Following that, there will be a detailed recommendation including a narrative recommendation, and a new system development.

II. Background Information

A. Brief Description of Project Request

The company would like for me to develop a system where the employees can quickly an accurately do their pump reconciliationis. The situation defined above is one that has way too many possibilities for various errors—errors increased drastically when the employees are either under trained, or simply do not have enough education to allow them to function effectively. They would be able to enter numbers into the computer, rather than having to compute them by hand—having to rely on a complicated system of entries where even something as insignificant as a badly written number can drastically effect your estimates of the amount fuel projected to be left in the underground tanks.

III. Detailed recommendation

A. Narrative recommendation

1. Quick Fixes

The most important quick fix will be to better train the employees. Many of them come in having either dropped out of High School, or simply are not educated enough to work with some of the complex numbers involved. They often simply mimic the actions of the person who trained them—not truly understanding what it is that they are doing and therefore causing more mistakes in the long run. Training should include more than just showing the employees how to function.

Rather, it should explain why things are done the way they are, as well as the implications of inaccurate readings or readings that don't measure up to the actual level of fuel in the tank. Perhaps a comprehensive test could be given right off and periodically to ensure that the employees are adequately trained.

2. New Systems Development

The system will be improved to keep track of sales per day, week, month, etc. This will help to determine whether or not one of the tanks are leaking and now much gasoline is lost due to people driving off without paying. If the reading taken from the gallons that have actually passed through the pumps do not match up with the actual number of gallons left in the underground tank then there is a leak in one of the tanks. On the other hand, if the gallons having passed through the pumps do not match the actual gallons sold by the cashier, then some of it is being either stolen, or not rung up at all by the cashier. The employees will be able to effectively input numbers and the program will do the rest.

B. Project plan

1. Initial Project Objectives

The initial project objective is to make sure that all problems with the old system of reconciling the pumps are met and dealt with. The forthcoming software must be able to adequately overcome all aspects of the reconciliation process, making it very important to completely understand the process. Once there exists a key understanding of the problem and problems at hand, then the software can be developed to take care of those problems.

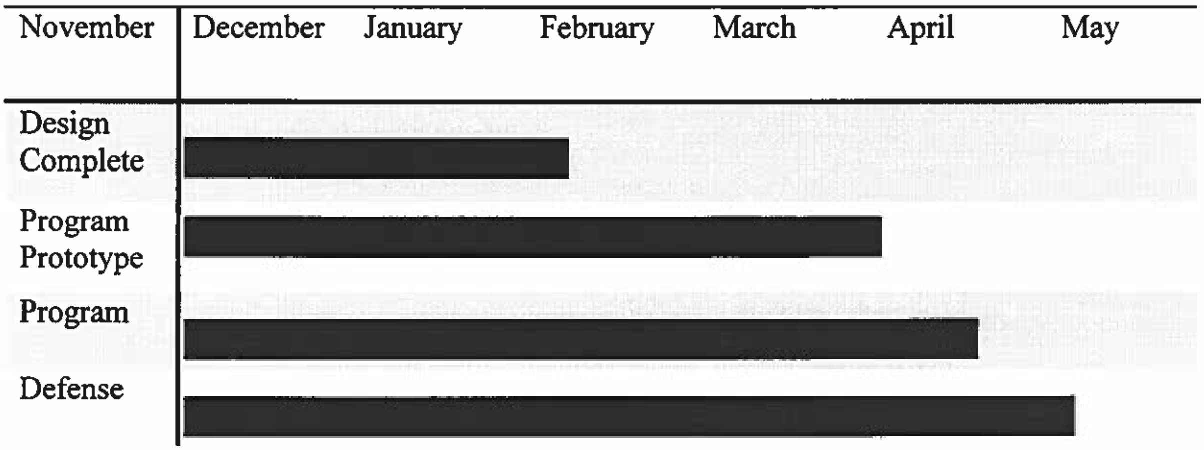
IV. Details

A. Variables Involved

- # Gallons left in the tanks
- # Gallons that have passed through the individual pumps

- Estimated fuel left in tanks
- Gallons sold
- Actual fuel left in tanks
- Equation for volume of cylinder
- Other (upcoming but not yet accounted for)
- Tank #
- Date

Time Table



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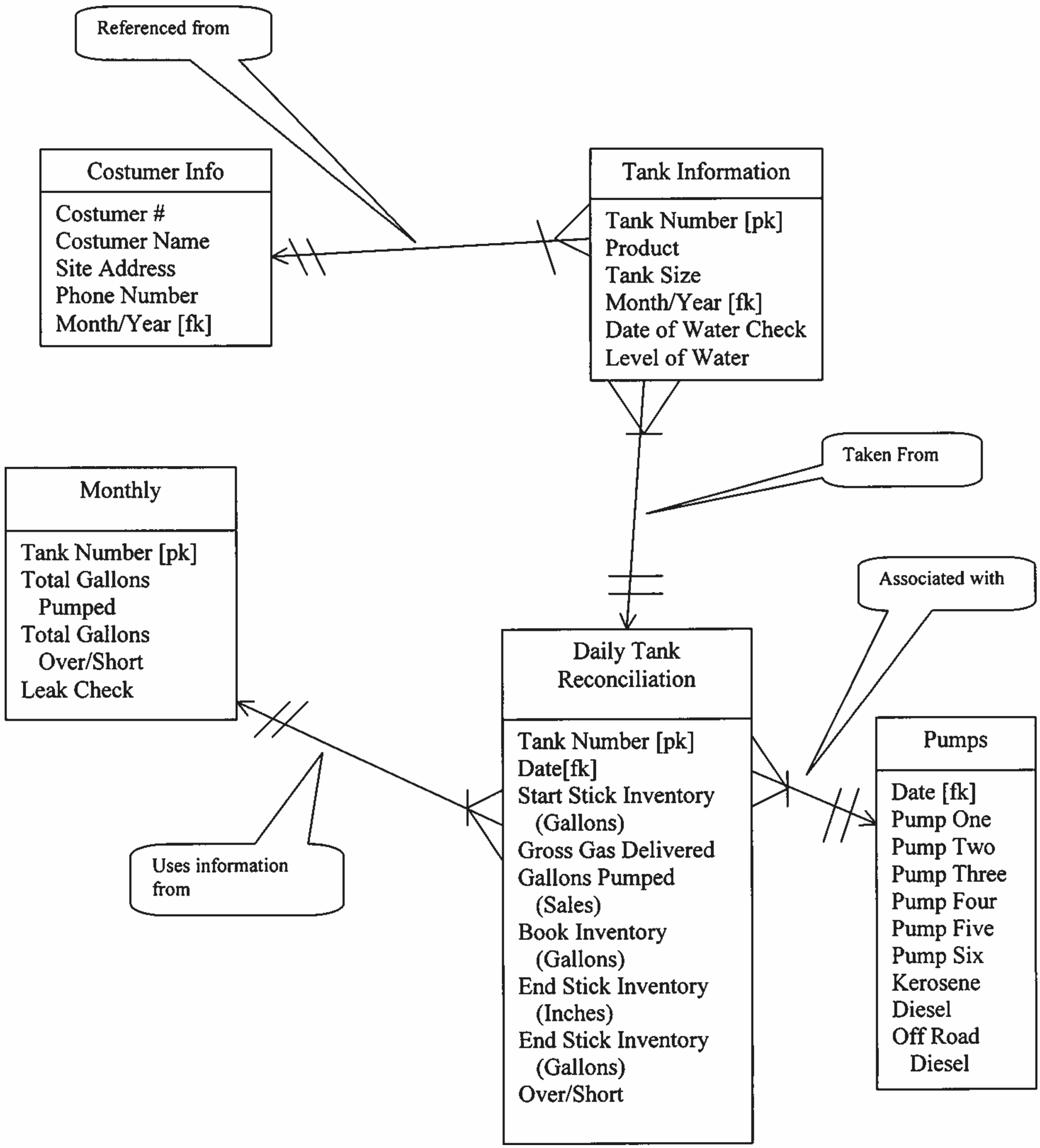
Problem Statements

Project: Pump Reconciliations	Project Manager: Christopher Corbett
Created by: Christopher Corbett	Last Updated By: Christopher Corbett
Date Created: 20 Jan 2002	Date Last Updated: 14-Apr-98

Brief Statements of Problem, Opportunity, or Directive	Urgency	Visibility	Annual Benefits	Priority or Rank	Proposed Solution
1. Current system does not account for ignorant employees.	ASAP	High	Will allow more flexibility in hiring, and allow for easier training and reduced mistakes.	1	Design a database where you can enter all of the information of automated.
2. The data needs to be stored in a more permanent and flexible way.	5 weeks	High	Will provide the owner with a much neater view of his records.	2	Allow the database to be as flexible as possible.
3. Possible water in the tanks, as well as tank leaks, need to be more carefully monitored.	4 weeks	High	Will let the owner be more thorough and accurate with his tanks.	3	Design the database to calculate the amount left in the tanks, and also make sure that the amount said to be left actually equals the amount in there—plus/minus 130 gallons.

Entities


Referenced from
Costumer Info
Costumer #
Costumer Name
Site Address
Phone Number
Month/Year [fk]
Tank Information
Tank Number [pk]
Product
Tank Size
Month/Year [fk]
Date of Water Check
Level of Water
Taken From
Monthly
Tank Number [pk]
Total Gallons Pumped
Total Gallons Over/Short
Leak Check
Associated with
Daily Tank Reconciliation
Tank Number [pk]
Date[fk]
Start Stick Inventory (Gallons)
Gross Gas Delivered
Gallons Pumped (Sales)
Book Inventory (Gallons)
End Stick Inventory (Inches)
End Stick Inventory (Gallons)
Over/Short
Pumps
Date [fk]
Pump One
Pump Two
Pump Three
Pump Four
Pump Five
Pump Six
Kerosene
Diesel
Off Road Diesel
Uses information from

Entity/Definition Matrix

ENTITY	BUSINESS DEFINITION
COSTUMER INFORMATION	The various unchanging information regarding the customer—such as name, address, ID #, etc
TANK INFORMATION	Specific information indigenous to a particular tank.
STICK MEASUREMENTS	Where the gas is measured manually to be compared to projected values.
PUMPS	One of the pumps that bring the fuel out of the ground, and keeps track of the number of gallons passing through.
MONTHLY	The monthly, routine jobs and calculations that must be preformed on the tanks.
DAILY TANK RECONCILIATIONS	The daily process of capturing the readings from the pumps and the tanks to see how many gallons have passed through the pump and how many gallons have actually left the tank.

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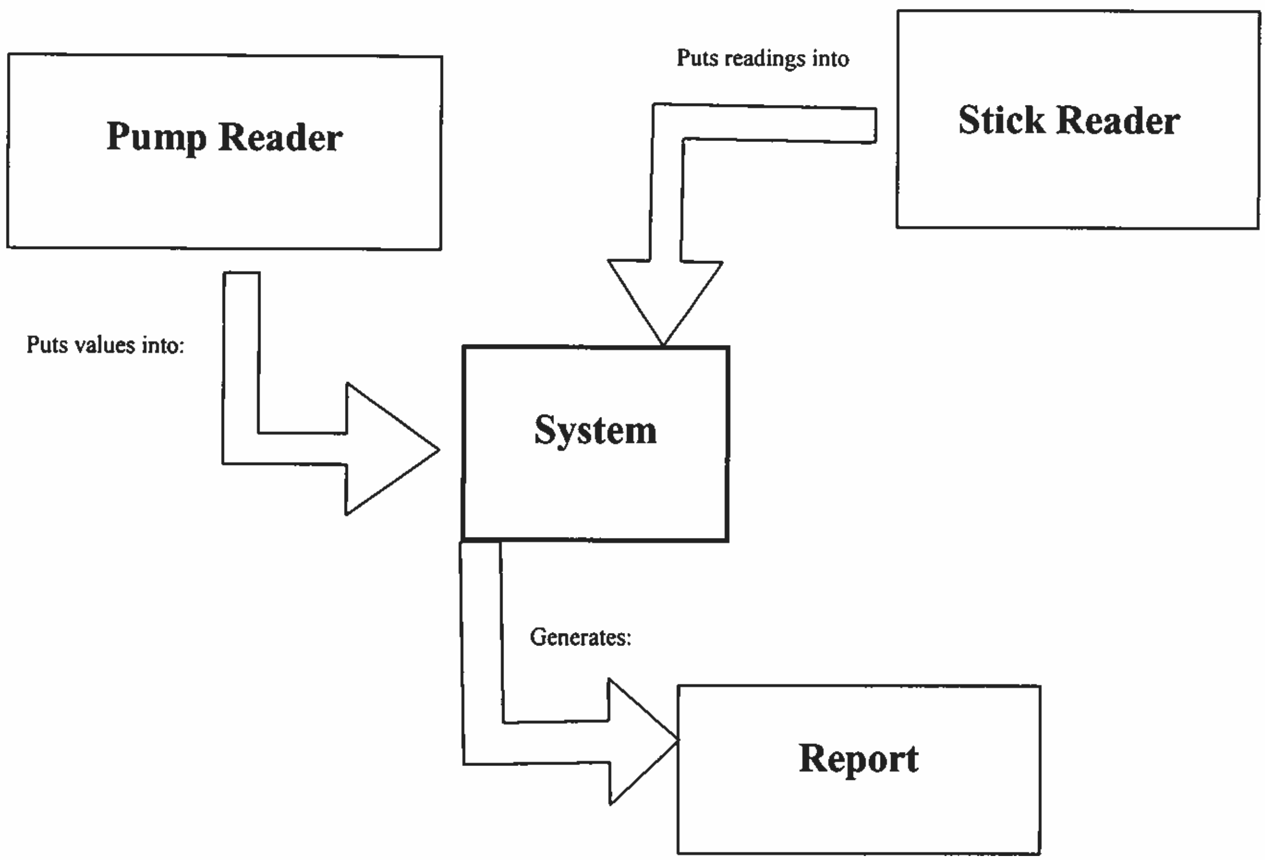
Context Diagram



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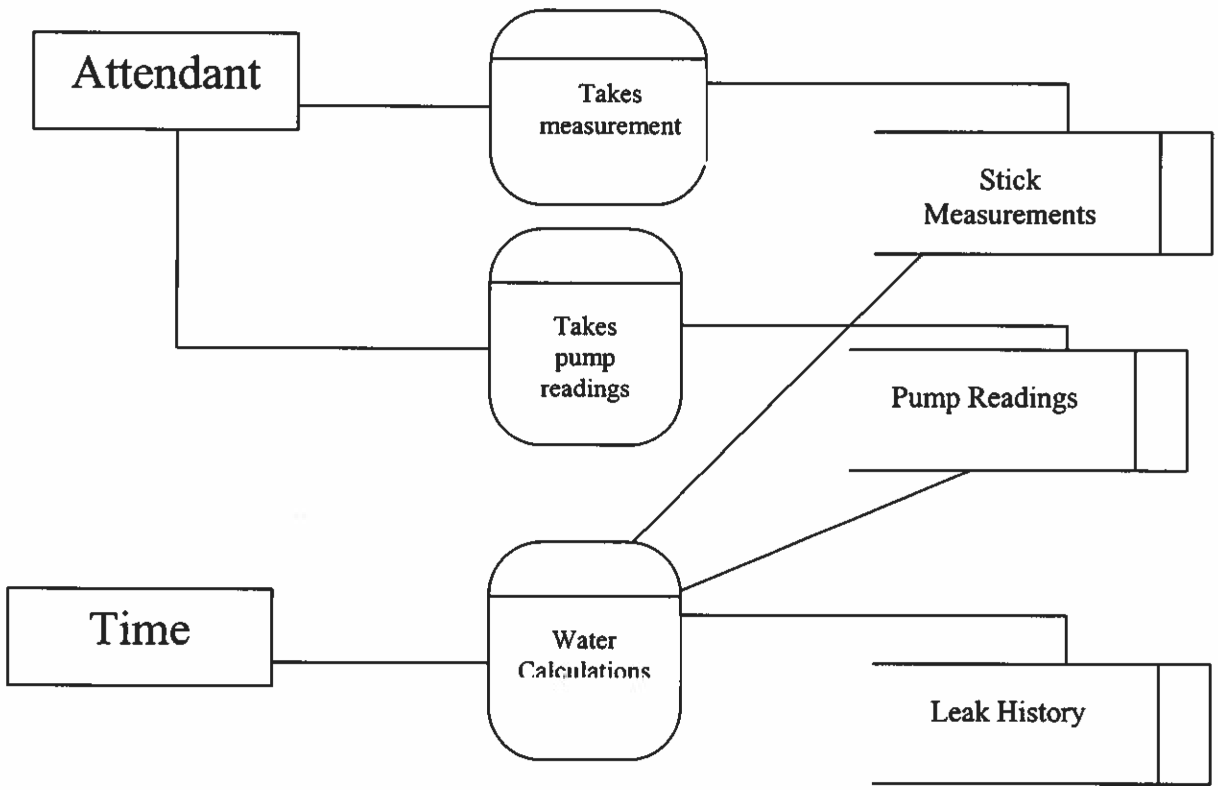
SYSTEM DIAGRAM



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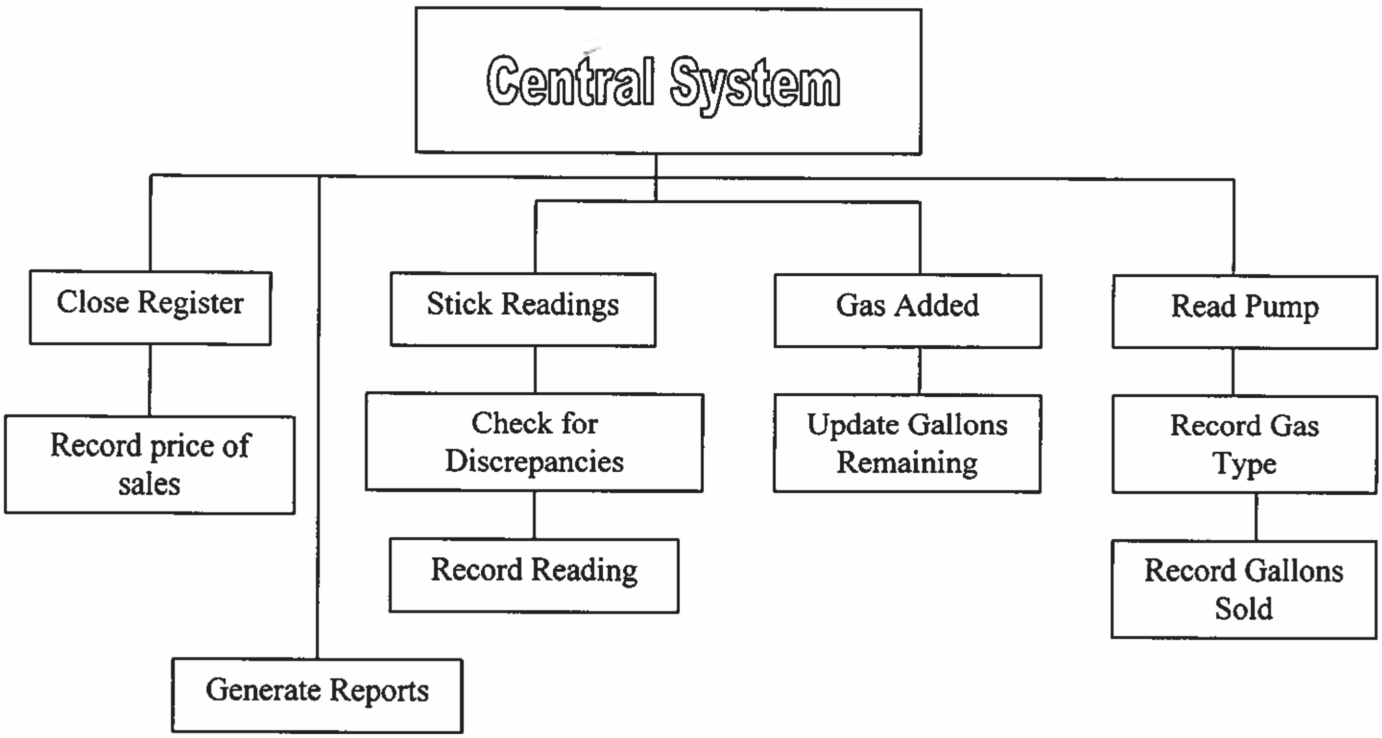
Event Decomposition Diagram



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EVENT LIST

Event Descriptions	Trigger (Inputs)	Responses (Outputs)
Gas added	Gas station calls in and orders more gasoline from distributor	Updates the amount of gasoline projected to be in tanks.
Gas taken out of tank	Someone stops at a pump and gets gas for their vehicle.	Projected amount in the tank is decreased.
Water found	Water is found in the gas tank.	User is alerted to possible water being in the tank, and takes action.
Drive off	Someone puts gas in their vehicle and never pays for it.	The user is alerted to the fact that a drive off occurred, and a record is made.
Tank leak	Projected amount in tank does not match the stick reading.	User is notified of the possible leak, and a record is kept.
Water check	Water check is initiated.	Record shows the water check.
Price change	The price that the station sells for changes.	The price for gallon is changed so that the records can be kept straight.

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COMPLETE TABLE DEFINITIONS

Table Name	Field Name	Type	Size	Description	Mask
Costumer Information	Costumer Number	Number		Long Integer	
	Costumer Name	Text	50		
	Site Address	Text	75		
	Phone Number	Text	50		
	Date	Date/Time		Short Date	
Daily Tank Reconciliations	Tank Number	Number		Long Integer	
	Date	Date/Time		Short Date	
	Start Stick Inventory (Gallons)	Number		Double	
	Gross Gallons Delivered	Number		Double	
	Gallons Pumped (Sales)	Number		Double	
	Book Inventory (Sales)	Number		Double	
	End Stick Inventory (Inches)	Number		Double	
	End Stick Inventory (Gallons)	Number		Double	
Monthly	Tank Number	Number		Long Integer	
	Total Gallons Pumped	Number		Double	
	Total Gallons Over/Short	Number		Long Integer	
	Leak Check	Number		Long Integer	
	Leak Found	Yes/No		Yes/No	
Pumps	Date	Date/Time		Short Date	
	Pump One	Number		Double	
	Pump Two	Number		Double	
	Pump Three	Number		Double	
	Pump Four	Number		Double	
	Pump Five	Number		Double	
	Pump Six	Number		Double	
	Kerosene	Number		Double	
	Diesel	Number		Double	
	Off Road Diesel	Number		Double	
Tank Information	Tank Number	Number		Long Integer	

	Product	Text	50		
	Tank Size	Number		Long Integer	
	Date	Date/Time		Short Date	
	Date of Water Check	Date/Time		Short Date	
	Level of Water	Number		Decimal	

Candidate Matrix

Characteristics	Candidate 1	Candidate2	Candidate 3
Portion of system Computerized Brief description of that portion of the system that would be computerized in this candidate.	General software designed to do some of the simple calculations and give daily printouts.	Moderately user friendly, and can keep some statistical analysis of the system and it's projections.	A system that could keep multiple variables stored, and output a variety of reports.
Benefits Brief description of the business benefits that would be realized for this candidate.	Would be a very basic system and therefore cost less.	Would be easier to use and also provide the user with a couple of visual charts.	Would be very user friendly, keep detailed records, and also provide a variety of charts and options.
Servers and Workstations Software tools needed to design and build the candidate (e.g., database management system, emulators, operating system, languages, etc.). Not generally applicable if applications software packages are to be purchased.	None required.	A processor that is at least 500 MHz, Windows 95 or higher.	A processor 800 MHz or better, Windows 2000 with GL support. This system would have to be highly efficient to support the graphics and the level of interface.
Software Tool Needed A description of the software to be purchased, built, accessed, or some combination of those techniques	No special tools required. Basic software package can be purchased online to use on a standard Windows® operating system.	Microsoft Access, Excel, both with Visual Basic support.	Microsoft Access, Excel, Visual Basic, Internet Support, Open GL programming.
Application Software A description of the software to be purchased, built, accessed, or some combination of these techniques.	Package Solution.	Custom Solution.	Advanced Custom Solution
Method of Data Processing Generally some combination of online batch, deferred batch, remote batch, and real-time.	Batch	Batch	Batch
Output Devices and Implications A description of output devices that would be used, special output requirements, (e.g. network preprinted forms, etc.), and output considerations (e.g., timing constraints).	Pre-drawn up templates using Microsoft Word.	Some customized reports formed within Access.	A variety of customized reports, to be generated within Access, also using Open GL for some comprehensive graphics.
Input Devices and Implications A description of input methods to be used, input devices (e.g., keyboard, mouse, etc.), special input requirements, (e.g. new revised forms from which data would be input), and input considerations (e.g., timing of actual inputs).	Keyboard, Mouse, Standard Desktop accessories.	Keyboard, Mouse, Standard Desktop accessories.	Keyboard, Mouse, Scanner, Standard Desktop accessories.
Storage Devices and Implications Brief description of what data would be stored, what data would be accessed from existed stores, what storage medias would be used, how much storage capacity would be needed, and how much data would be organized.	Local Hard Disk.	Local Hard Disk.	Local Hard Disk.

Feasability Matrix

Feasibility Criteria	Wt.	Candidate 1	Candidate 2	Candidate 3
Operational Feasibility **Functionality.** A description of to what degree the candidate would benefit the organization and how well the system would work. **Political.** A description of how well received this solution would be from user management, user, and organization perspective.	60%	Pre-packaged software from the market, not specific to the company's needs. Score: 60	Would support the periodic evaluations and some statistical information by analyzing and compiling them into spreadsheets. Score: 85	Same as candidate 2, but would also include more statistical options, as well as a graphical interface for more a comprehensive understanding. Score: 100
Technical Feasibility **Technology.** An assessment of the maturity, availability (or ability to acquire), and desirability of the computer technology needed to support this candidate. **Expertise.** An assessment of the technical expertise needed to develop, operate, and maintain the candidate system.	20%	Could be upgraded as the designers produce updated versions. This package would require that the operators be somewhat technically efficient. Score: 65	Could be upgraded upon demand or by any other programmer with Access and Excel experience. Score: 100	Same as Candidate 2, but would require the programmer to be familiar with Open GL, some Java, and various html formats. Would probably be costly to upgrade. Score: 80
Economic Feasibility **Cost to develop:** **Payback period (discounted):** **Net present value:** **Detailed calculations:**	0%	Approximately $ Approximately $ Approximately $ Score: n/a	Approximately $ Approximately $ Approximately $ Score: n/a	Approximately $ Approximately $ Approximately $ Score: n/a
Schedule Feasibility An assessment of how long the solution will take to design and implement.	20%	Less than a month Score: 100	Less than 3 months Score: 100	Over 9 months Score: 70
Ranking.	**100%**	225 (75%)	285 (95%)	250 (83%)

SYSTEM PROPOSAL

I. Introduction

A. Purpose of the report

The purpose of this report is to present the user with a written discussion of the proposals listed in the Candidate Matrix. I will attempt to better explain the various solutions from a professional perspective, but the final choice remains with the company as to which candidate they would rather implement.

B. Background of the project leading to this report

To date, many man hours have been spent analyzing the current system of pump reconciliations and I have come up with three choice candidates. Any of those three choices should be sufficient in helping you to continue carrying out your daily operations. Some of them are more versatile than others, but the most versatile and more tailored ones always have price tradeoffs.

C. Scope of the project

The scope of the project is to create a system that will allow employees with little education to complete the rather detailed pump reconciliation charts. IT will also help to prepare charts and efficient print-outs either daily or periodically.

D. Structure of the report

This report will give a recommendation based on my professional view of what would best encompass the needs of the company. The factors in my evaluation range from the price, the versatility, the degree to which a particular candidate will be tailored to the system, and the obvious needs of the company and individuals involved.

II. Tool and techniques used

A. Solution Generated

After carefully reviewing all of the information presented, my personal recommendation is that the company consider the second candidate as solution. This candidate corresponds best with the desired outcome in terms of both cost and efficiency. The most influential reasons in this decision come down to cost and performance vs. practicality. The first candidate would be barely sufficient, and the third candidate would most certainly be an overkill.

B. Feasibility analysis (cost-benefit)

The current computer would be adequate for the software needed in the new system. The third candidate requires the company, with a limited budget, to purchase all software need to implement the system along with a very expensive machine to run it. The graphics and the power of the computer in the third candidate are nice, but would not be practical for a project of this scale. The cost-benefit would certainly favor the second candidate.

III. Information system requirements

This second candidate would require the user to purchase copies of Microsoft Access®, as well as Microsoft Visual Basic ®. The system that will be designed will be very simple to operate and also be very simple to maintain. I will be the one designing it—which benefits the both of us when it comes to terms of implementation and training, because I'll be on hand for all your questions.

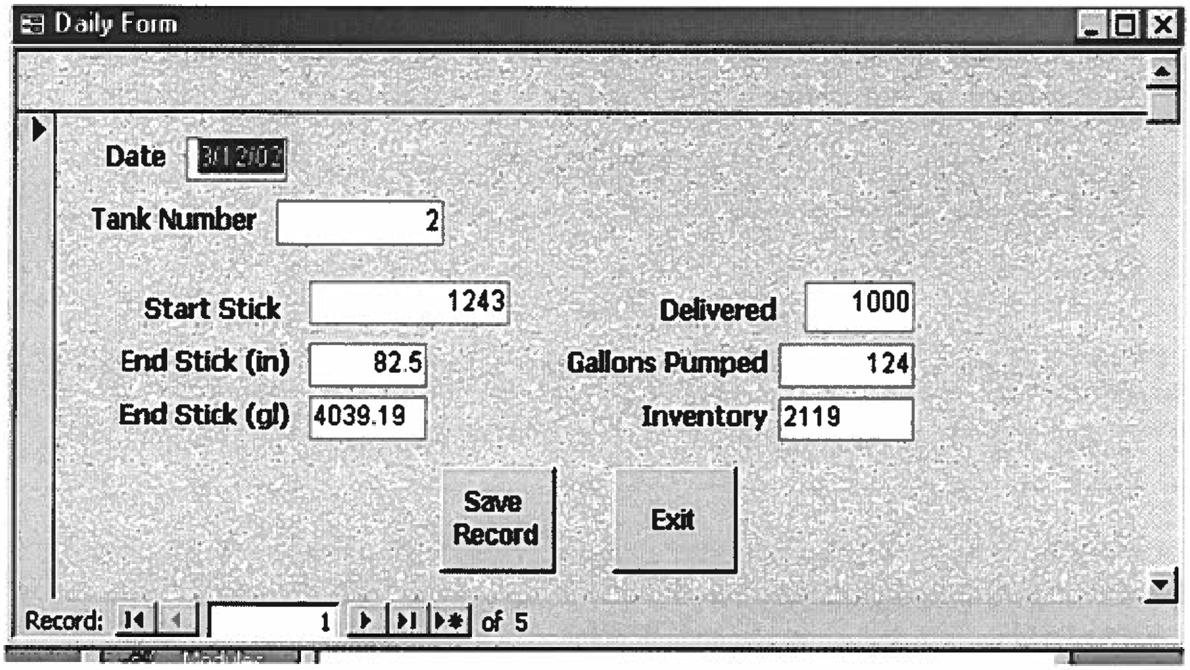
IV. Alternative solutions and feasibility analysis

The other solutions are obviously candidates one and three. The first candidate is nothing at all fancy. It would use Microsoft Word's ability implement various features of Excel, Visual Basic, and Access functions. The solution would be designed as a template that the data could be entered every day and then printed up and either saved onto the hard drive, or stored on a hard copy.

The third candidate would be effective, but also very expensive. It would constitute taking Visual Basic and compiling it into a database of it's own, able to pull from graphics programs such as Visual Basic, linking it across multiple databases, and also directly accessible over the internet. The program could be compiled and loaded onto any machine as a complete program—independent of other Microsoft applications.

V. Recommendations

Considering all of the information presented, I am recommending Candidate 2 to be designed and implemented as soon as possible. It's features fit very well into the scope of the project request, and it fits nicely into a small budget and a fast design phase and implementation.



This form is the daily form [database side] that the user will be interacting with a lot.

The inventory is calculated by :
Start Stick [gl] + Delivered [gl] – Gallons pumped.

The Code for Save Record and Exit is written in Visual Basic and is as follows:

```
Option Compare Database

Private Sub Command4_Click()
'On Error GoTo Err_Command4_Click

   DoCmd.GoToRecord , , acNewRec

Exit_Command4_Click:
   Exit Sub

Err_Command4_Click:
   MsgBox Err.Description
   Resume Exit_Command4_Click

End Sub
```

```
Private Sub Command5_Click()
'On Error GoTo Err_Command5_Click


   DoCmd.Close

Exit_Command5_Click:
   Exit Sub

Err_Command5_Click:
   MsgBox Err.Description
   Resume Exit_Command5_Click

End Sub
```

Daily Update

Date	Tank Number	Start Stick	Delivered	Gallons Pumped	Book Inventory	End Stick (in)	End Stick (gl)
3/12/02	2	1243	1000	124	0	82.5	0
3/13/02	4	4560	0	535	0	83	0
3/14/02	2	124	120	135	1331	344	1234
4/21/02	3	1500	1000	120	0	48.6	0
4/22/02	5	4500	500	300	0	96	0

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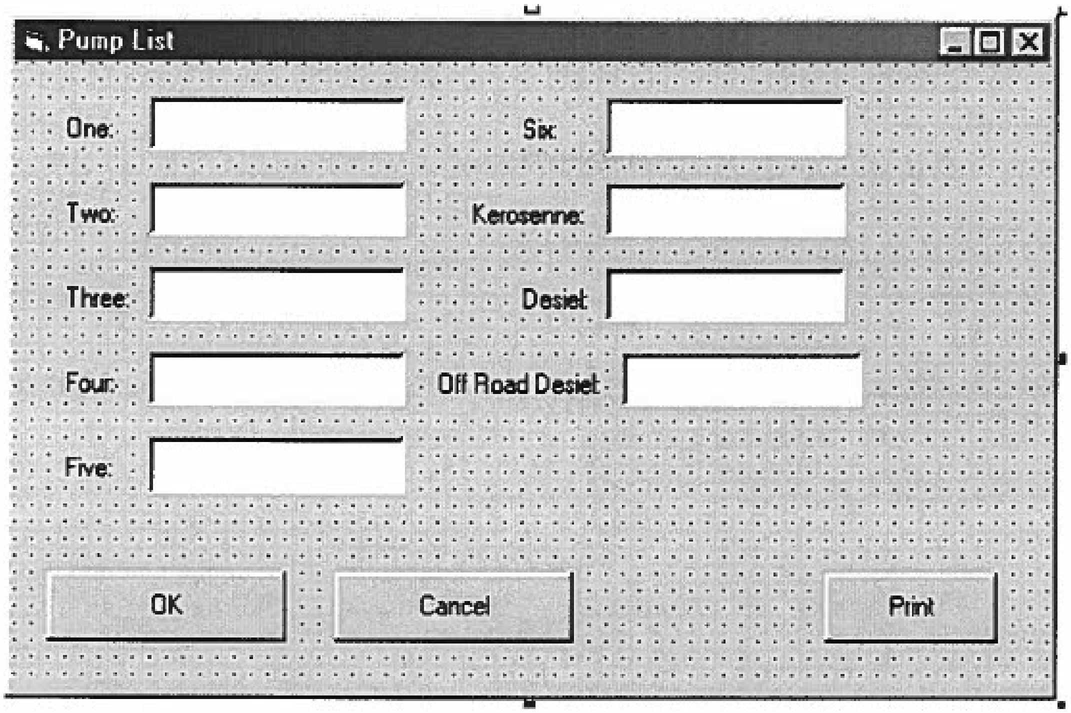
Here is an example of how a daily printout could be generated and printed quite easily. This particular printout would be a daily hard copy of all of the gasoline transactions in the store.

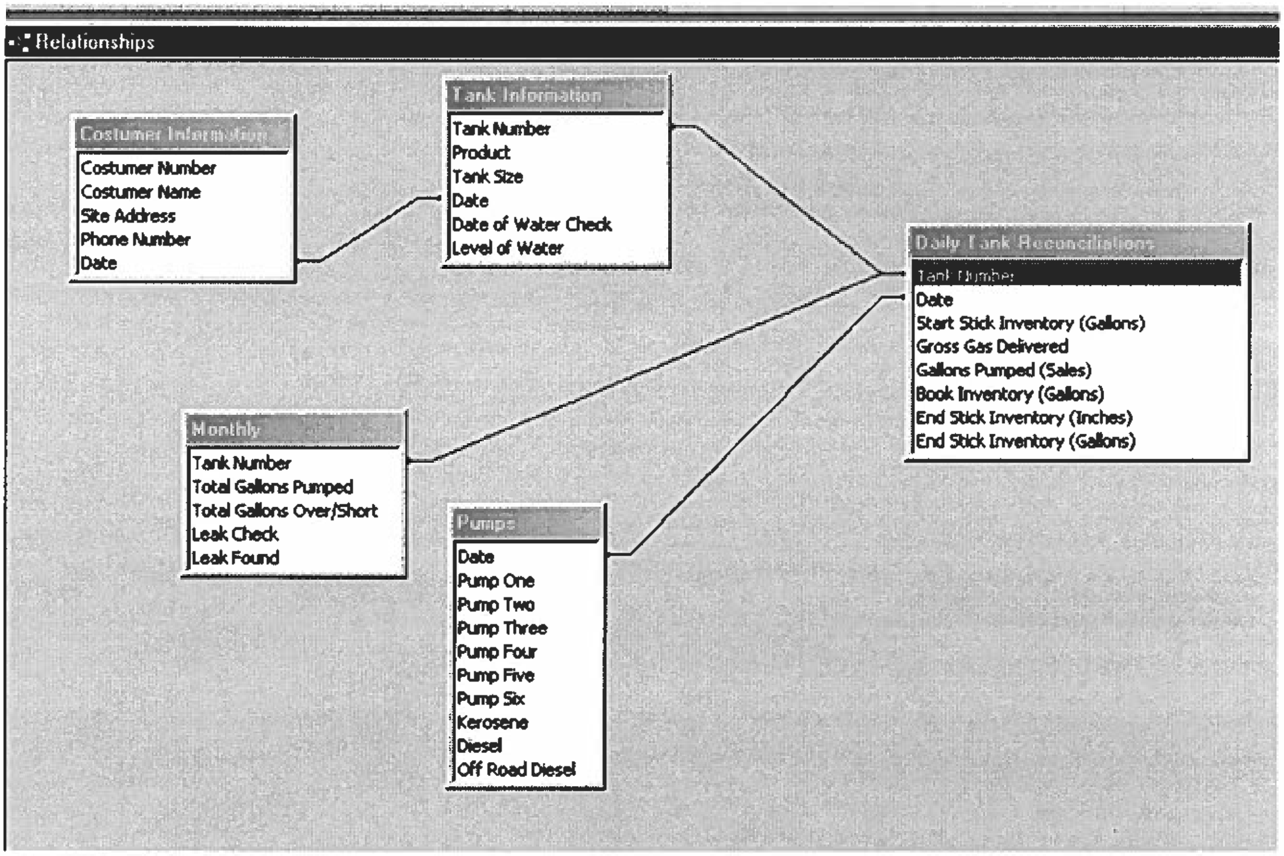


From this form, you can enter the daily values from the pump and it will be stored in the Database.

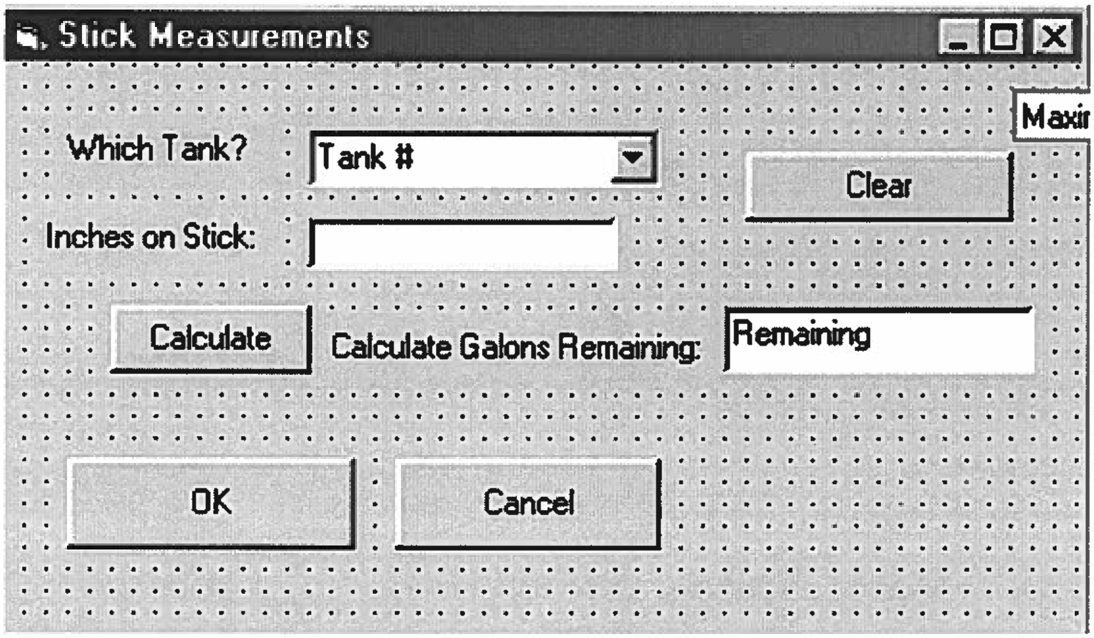
The Ok button will capture and save all the values. Upon linking the program with Microsoft Access, the values will simply fall under the record for whatever date entered under the Main Menu.

The Cancel button will go back to the main screen without saving.

The Print button will print off the entries for a daily record sheet.

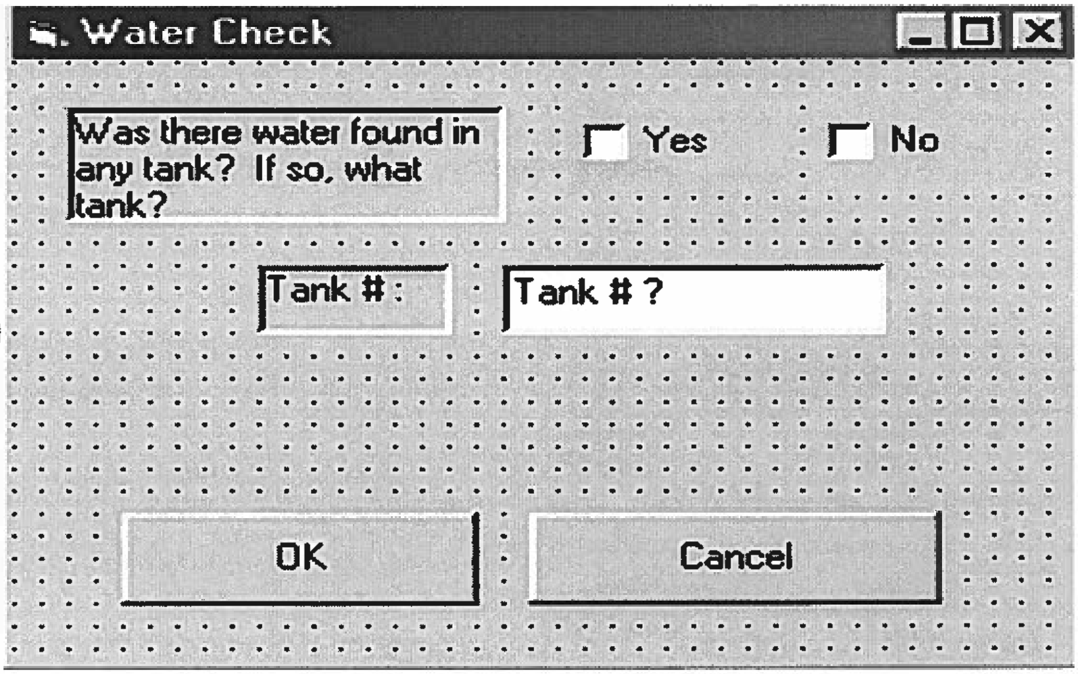


This is a diagram of all of my entities and the relationships that they fall under. My Entity chart is more comprehensive, but this should give you an immediate view of how everything is connected together.



This is the form that the stick measurements are entered from. There are six different tanks, three of the having different volumes. When he tank is selected, it's dimensions are put into an equation along with the stick reading, and then the Calculate button will calculate the gallons remaining.

The Ok button saves everything and goes back to the main menu, and the Cancel button simply returns to the main menu without saving anything.



This is the Visual Basic user interface for the water check. The date, as recalled Is pre-entered in the Main screen. This simple form allows the user to enter whether or not he completed a water check, and also whether any water was found.

CODE FROM:
Database → Forms →Daily Form

```
Option Compare Database

Private Sub Command4_Click()
'On Error GoTo Err_Command4_Click


    DoCmd.GoToRecord , , acNewRec

Exit_Command4_Click:
    Exit Sub

Err_Command4_Click:
    MsgBox Err.Description
    Resume Exit_Command4_Click

End Sub
Private Sub Command5_Click()
'On Error GoTo Err_Command5_Click


    DoCmd.Close

Exit_Command5_Click:
    Exit Sub

Err_Command5_Click:
    MsgBox Err.Description
    Resume Exit_Command5_Click

End Sub
```

CODE FROM
Visual Basic → Senior Project → frm.Main (Main Form)

```
Private Sub btnOkay_Click()

If cmdJob = "Water Check" Then frmMain.Visible = False
frmWater.Visible = True
frmMain.Visible = False
frmPumps.Visible = False
frmStick.Visible = False

If cmdJob = "Pump Inventory" Then frmMain.Visible = False
frmPumps.Visible = True
frmMain.Visible = False
frmStick.Visible = False
frmWater.Visible = False

If cmdJob = "Stick Measurements" Then frmMain.Visible = False
frmStick.Visible = True
frmMain.Visible = False
frmPumps.Visible = False
frmWater.Visible = False


End Sub
```

CODE FROM:
Visual Basic → Senior Project → frmStick

```
Private Sub btnCalculate_Click()
 ' The following is the formula for the volume of a
 ' Cylinder laying on its side:
 '
 '                  (r - y)
 '    V = h[r^2) (arccos -----)( - (r - y)*sqrt(2ry - y^2)]
 '                     r


Dim r#  ' Radius of the tank
Dim y#  ' Level form the bottom of the tank
Dim h#  ' Length of the tank
Dim V#  ' Volume of the tank
r = 6
h = 14
y = Val(txtInches.Text)
V = (h * (r ^ 2) * ((1 / Cos((r - y) / (r))))) - ((r - y) * Sqr(2 * r * y - y ^ 2))
txtRemaining.Text = V


End Sub

Private Sub btnCancel_Click()
frmStick.Visible = False
frmMain.Visible = True
End Sub

Private Sub btnClear_Click()
txtRemaining.Text = " "
txtInches.Text = " "
End Sub

Private Sub btnOk_Click()
txtRemaining.Text = " "
txtInches.Text = " "
frmStick.Visible = False
frmMain.Visible = True
End Sub
```